

The Clebsch hexagon code: rigidity of a conic syndrome locus

An MDS code over $\mathbb{F}_{11}$ whose projective deep-hole directions form a conic

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Abstract

Among the six-arcs of $\text{PG}(2, 11)$, exactly one projective class has the maximum-distance syndrome directions of its associated MDS code contained in a conic: the Clebsch hexagon. This coding-theoretic condition recovers its projective stabilizer A_5 , and the characterization remains true with the zero set of any nonzero form of degree at most three in place of a conic.

The arc lies on no conic, so its $[6, 3, 4]_{11}$ MDS code is non-GRS and has covering radius three. Its twelve projective deep-hole directions are the $\mathbb{F}_{11}$ -points of a conic; they lift to 120 deep-hole cosets, 2400 minimum-weight leaders, and 159720 received-word deep holes. A quadratic syndrome test is a complete distance oracle, and its ambiguity data recover the classical Brianchon points and self-polar triangles.

We prove a five-way rigidity equivalence and a sharp nearest-conic gap of twelve, decompose the 252 one-point perturbations into eight A_5 -orbits, and construct an invariant unordered $10 + 10$ bipartition of the twenty weight-three support types. A chord-defect identity and the Sylvester graph show that $q = 11$ is the only conic-filling field for six-arcs. For arc sizes four through seven, the only conic-filling cases are a four-frame in $\text{PG}(2, 5)$ and the Clebsch hexagon. The six-arc census is classical; the new content is its geometric and coding-theoretic rigidity, quantitative gap, decoding structure, and finite-field boundary.

Keywords. Clebsch hexagon; MDS code; deep hole; finite projective plane; arc; conic.

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1 Introduction

Let C be a linear $[n, k, d]_q$ code with covering radius ρ and full-rank parity-check matrix H . For a received word v , write $\sigma(v) = Hv^\top$ and

$$w(s) = \min\{\text{wt}(e) : He^\top = s\}.$$

Then $d(v, C) = w(\sigma(v))$, and v is a *deep hole* precisely when this minimum is ρ . The syndrome s specifies one coset of C , whose coset leaders are its weight- $w(s)$ representatives. Because $w(\lambda s) = w(s)$ for $\lambda \neq 0$, maximum-distance syndrome directions form a projective locus

$$\mathcal{D}_{\text{proj}}(C) = \{[s] \in \text{PG}(n - k - 1, q) : w(s) = \rho\}.$$

Thus one projective direction represents $q - 1$ distinct nonzero syndrome cosets, while each coset contains $|C|$ received words.

For an MDS code of redundancy $r = n - k$, the parity-check columns form a projective arc in $\text{PG}(r - 1, q)$, and $w(s)$ is the least number of columns whose span contains $[s]$. Adjoining a new column preserves the MDS property exactly when its direction avoids

every hyperplane spanned by $r - 1$ existing columns. In the redundancy-three plane case used here, this specializes to weights one, two, or three according as the direction lies on the arc, on a secant of the arc, or on neither; an extension point is precisely a direction missed by every secant. This arc–coset dictionary, in various forms, underlies the classical theory of complete arcs and saturating sets and has been made precise and general for MDS codes by Davydov, Marcugini, and Pambianco [9], whose Theorem 6.3 gives the exact leader-count formula we use in Section 3. For plane parity-check systems — equivalently redundancy-three $[n, n - 3, 4]_q$ MDS codes — the same dictionary is implicit already in the classical arc/covering-radius literature; see Hirschfeld [19]. We write $\mathcal{U}(A)$ for this projective extension locus, the directions off A and off all its secants. It is the weight-three syndrome-direction locus. When $\mathcal{U}(A) \neq \emptyset$ the associated redundancy-three code has $\rho = 3$ and $\mathcal{U}(A) = \mathcal{D}_{\text{proj}}(C)$; when $\mathcal{U}(A) = \emptyset$ the covering radius is at most two, and its deep-hole directions are therefore not described by $\mathcal{U}(A)$. For the standard (affine) Reed–Solomon setting, determining deep holes is a long-studied algorithmic and structural question; Guruswami and Vardy [18] initiated its decision-complexity study, showing maximum-likelihood decoding of Reed–Solomon codes to be NP-hard by essentially proving that deciding whether a received word is a deep hole is co-NP-complete, and Cheng and Murray [8] gave a substantially simpler proof of that same co-NP-completeness result and extended the study to standard (full-evaluation-set) Reed–Solomon codes. Zhang, Wan, and Kaipa [30] determine the deep holes of *projective* Reed–Solomon codes completely, identifying them with tangent lines to, and a quadratic-extension family over, the underlying normal rational curve (their Theorems I.4–I.7). Both of these are for the generalized Reed–Solomon (GRS) case, where the arc lies on a normal rational curve; the present code does not.

The object of this paper is a single, small, and highly symmetric exception to that GRS world: a $[6, 3, 4]_{11}$ MDS code not monomially equivalent to a GRS code, whose covering radius is three and whose projective deep-hole syndrome locus — the directions missed by the arc’s fifteen secants — coincides exactly with the $\mathbb{F}_{11}$ -points of a single nonsingular conic. The arc realizing this code is the *Clebsch hexagon*: six points of $\text{PG}(2, 11)$. Its real-projective antecedent goes back to Clebsch [4, p. 336]; Edge constructed the finite-plane configuration, named it a Clebsch hexagon, and determined its order-60 stabilizer [5, §§29–32]. Dye developed its synthetic geometry over a general field, proved its projective uniqueness, and determined its stabilizer [16, Theorems 1 and 3, pp. 275–278]; Storme and Van Maldeghem identified the corresponding six-arc in their finite-plane classification [27].

The main theorem starts from no symmetry or conic construction. Among all six-arcs of $\text{PG}(2, 11)$, conic containment of the projective deep-hole syndrome locus reconstructs the Clebsch hexagon and its A_5 stabilizer. A line bound, a chord-defect identity, and Dye’s Brianchon theorem give the conceptual implication; the census is needed only for the numerical clause. We then prove global and local gaps, an invariant chirality split of the leaders, decoder reconstruction of the Brianchon geometry, and the uniqueness of $q = 11$. For the Clebsch family the same chord identity gives $|\mathcal{U}(H)| = q^2 - 14q + 45$, with off-conic excess $(q - 4)(q - 11)$ when $q \equiv 3 \pmod{4}$.

What is new and what is not. The hexagon, its uniqueness and stabilizer, and its complete-exterior-set interpretation are classical [4, 5, 16, 27, 2]. Sadeh classified the six-arcs of $\text{PG}(2, 11)$ [24]; we claim no priority for that census or its extension counts. The companion arcs paper records the single conic-locus identification of Proposition 3.2, with a kernel-checked certificate [7, Prop. 8.7]; we reprove it for self-containment. The new contribution is the symmetry-free coding criterion and its consequences: low-degree rigidity, quantitative gaps, leader chirality, decoder reconstruction, and the finite-field boundary.

2 The Clebsch hexagon

Fix the nonsingular conic

$$\mathcal{C} : XZ = Y^2$$

in $\text{PG}(2, 11)$, with its 12 points identified with $\text{PG}(1, 11) = \mathbb{F}_{11} \cup \{\infty\}$ in the usual way. The stabilizer of $\mathcal{C}$ in $\text{PGL}(3, 11)$ is $\text{PGL}_2(11)$, acting on $\mathcal{C}$ as it acts on the projective line.

Definition 2.1 (Clebsch hexagon). Let $A_5 \subset \text{PGL}_2(11)$ be an icosahedral subgroup. The *Clebsch hexagon* is the six-point orbit of poles of the six 5-fold-axis chords of A_5 acting on $\mathcal{C}$; equivalently, the six points of $\text{PG}(2, 11)$ each of whose two tangency lines to $\mathcal{C}$ meet $\mathcal{C}$ in an antipodal pair fixed by a Sylow 5-subgroup of A_5 .

For a nonsingular conic over an odd field, an *external point* lies on two tangents, while an *external line* is disjoint from the conic. Edge constructs exactly this $q = 11$ hexagon: its six vertices are external points, all fifteen joins are external lines, and the joins concur three at a time at ten internal Brianchon points [5, §§29–32]. In the terminology of Blokhuis, Seress, and Wilbrink, it is the six-point complete exterior set of $\mathcal{C}$ [2, pp. 143, 146]; see Van de Voorde [28] for the later exterior-set context. This exterior-set condition says only that the joins avoid $\mathcal{C}$. Proposition 3.2 is the strictly stronger covering statement that those joins cover every point outside the hexagon and $\mathcal{C}$. Van de Voorde also relates the broader sets-without-tangents problem to stopping sets in projective-plane LDPC codes; that is a different coding connection from the MDS covering-radius and deep-hole interpretation developed here.

The arc hypothesis is essential even inside this classical exterior-set category. Blokhuis–Seress–Wilbrink’s exhaustive $q = 11$ list has two complete exterior configurations: the Clebsch six-arc and a Pasch configuration [2, p. 146]. The latter has four collinear triples on its six points, so it cannot be the column set of a redundancy-three MDS code. Thus complete exteriority by itself gives only the inclusion $\mathcal{C}(\mathbb{F}_{11}) \subseteq \mathcal{U}(A)$; imposing the arc/MDS condition selects the Clebsch branch on which the exact covering and rigidity questions of this paper are posed.

There is also a more direct fixed-conic binary-code lineage. Droms, Mellinger, and Meyer formed LDPC codes from the four internal/external and passant/secant point–line incidence matrices [17]; two of their dimension conjectures—the secant–external and passant–internal cases—were subsequently proved in [25, 21]. Wu then used whole $\text{PSL}_2(q)$ -orbit families of conics contained in the internal-point set, and Madison and Wu treated the external-point analogue [29, 22]. These are binary incidence codes whose coordinates range over an entire internal or external point class. For example, at $q = 11$ the passant–internal null-space code has parameters $[55, 25]_2$, while Wu’s internal-orbit-conic null-space code has parameters $[55, 31]_2$. They are therefore different code objects from the six-coordinate $\mathbb{F}_{11}$ -linear MDS code studied here; the shared conic, polarity, and $\text{PSL}_2(11)$ geometry is infrastructure rather than a competing covering-radius or deep-hole identification.

This is projectively the arc K_2 of Storme and Van Maldeghem [27, Prop. 11] at $q = 11$, and the hexagon studied synthetically by Dye under the existence condition 5 a square in the base field [16, Theorem 1]. Dye’s Theorems 1 and 3 give projective uniqueness and stabilizer A_5 directly; Storme and Van Maldeghem’s Proposition 12 gives the corresponding finite-plane catalogue statement. Dye’s unique-polarity theorem identifies an associated conic; in our chosen coordinates the directly verified conic is $\mathcal{C}$. His calculation also shows that the six vertices lie on no conic in characteristic 11 [16, Theorem 2, pp. 277–278; p. 281].¹ The

¹The primary texts separate two statements that are easy to conflate. Dye proves that, over $\mathbb{F}_{11}$, every edge is external to the associated conic [16, discussion preceding Theorem 6, p. 281], and Blokhuis–Seress–

arcs of $\text{PG}(n, q)$ fixed by a *primitive* A_5 or A_6 were catalogued more generally by O’Keefe and Storme [20], of which the present arc is the planar A_5 case (its stabilizer is 2-transitive on the six points, hence primitive); we cite that catalogue for completeness, and take the planar uniqueness statement we use from [27, Prop. 12]. Storme and Van Maldeghem’s Proposition 13 records, by computer search, that this six-arc is incomplete at $q = 11$ (unlike at $q = 9$, where the analogous A_5 -hexagon is complete); equivalently, its associated redundancy-three code has covering radius exactly three. Incompleteness gives the lower bound, and the three-dimensional syndrome space gives the upper bound.

We work throughout with the explicit representative

$$A = \{(1, 10, 0), (1, 9, 1), (1, 4, 7), (1, 8, 5), (0, 1, 4), (1, 1, 7)\},$$

already used in the companion arcs manuscript [7] for the same reasons: it is disjoint from $\mathcal{C}$, no three of its points are collinear, and its stabilizer in $\text{PGL}(3, 11)$ has order 60.

3 The code and its projective syndrome locus

Form the 3×6 matrix H whose columns are (representatives of) the six points of A , and let

$$C = \{c \in \mathbb{F}_{11}^6 : Hc^T = 0\}$$

be the resulting $[6, 3, 4]_{11}$ MDS code. Since the six columns of H do not lie on any conic (their evaluation matrix against the six quadratic monomials $X^2, XY, XZ, Y^2, YZ, Z^2$ is nonsingular), C is not monomially equivalent to a GRS code: the dual of a GRS code is GRS, and the parity-check column system of a length-six, dimension-three GRS code lies on a normal rational curve, here a conic, in the classical normal-rational-curve/GRS dictionary [26].

By the arc–coset dictionary [9, Def. 6.1, Thm. 6.3], for every nonzero syndrome s above a projective direction $x = [s]$, the corresponding coset has minimum weight one, two, or three according as $x \in A$, x lies on a secant of A , or x lies on neither. In the last case each of the $q - 1$ cosets above x has exactly $\binom{n}{3}$ weight-three leaders, here $\binom{6}{3} = 20$. Their Theorem 6.3 is stated for an arbitrary n -arc of $\text{PG}(2, q)$ and its associated $[n, n - 3, 4]_q$ MDS code, with no Reed–Solomon, normal-rational-curve, or conic hypothesis and no restriction on q or the characteristic; it therefore applies to the non-GRS code C at hand. (The GRS and conic specializations appear separately in their Theorems 6.4, 6.5, 6.8, and 6.10, which we do not use. The leader count is also the $d = 4$, $n = 6$ case of the general $\binom{n}{d-1}$ of [9, Thm. 7.7], so it is an instantiation of a uniform formula rather than a coincidence at these parameters.) Finally, by their Definition 6.1 together with the definition of arc completeness, adjoining x as a seventh arc column preserves the MDS property exactly when x is missed by every secant.

The stabilizer already forces the small point sets that will occur.

Proposition 3.1 (The A_5 point orbits). *Let $G = \text{Stab}(A) \cong A_5$. Its orbits on $\text{PG}(2, 11)$ have lengths*

$$6, 10, 12, 15, 30, 30, 30.$$

The first four are respectively A , the ten Brianchon points, $\mathcal{C}(\mathbb{F}_{11})$, and the fifteen vertices of the five self-polar triangles. In particular, $\mathcal{C}(\mathbb{F}_{11})$ is the unique twelve-point G -orbit.

Wilbrink call the resulting six-arc a complete exterior set; thus the conic is contained in the uncovered locus. Neither source states the converse covering assertion that every off-conic point lies on an edge, which is the equality in Proposition 3.2.

Proof. Dye’s polarity and stabilizer theorems identify G with the ternary icosahedral representation preserving $\mathcal{C}$, and his Corollary 1 makes it transitive on the six vertices, ten Brianchon points, and fifteen triangle vertices [16, Theorems 2–3 and Corollary 1, pp. 276–279]. For completeness, we derive the orbit sizes rather than import them from the finite enumeration.

Since $11 \nmid 60$, the representation is semisimple, and the ternary icosahedral representation is irreducible; hence G fixes no projective point. The ordinary three-dimensional icosahedral character, reduced in $\mathbb{F}_{11}$, shows that an involution, an element of order three, and an element of order five fix respectively 13, 1, and 3 projective points. Indeed, an involution has a one-dimensional eigenspace and a two-dimensional eigenspace; the nontrivial order-three eigenvalues are not in $\mathbb{F}_{11}$; and the three order-five eigenvalues are distinct because $5 \mid 10$. Restricting the same representation to the standard subgroups of A_5 gives the following common-fixed-point ledger; the standard subgroup classification of A_5 shows that these are all possible nontrivial proper point stabilizers. Here D_5 has order ten, and the last column counts points whose stabilizer is exactly the displayed subgroup type.

H	A_4	D_5	S_3	C_5	V_4	C_3
$\#H$	5	6	10	6	5	10
$ \text{Fix}_{\text{PG}}(H) $	0	1	1	3	3	1
exact-stabilizer points	0	6	10	12	15	0

To spell out the subtraction in the last row: the normalizer D_5 fixes one of the three C_5 eigenlines and interchanges the other two, giving 6 points with stabilizer D_5 and 12 with stabilizer C_5 ; the unique C_3 -fixed line is already fixed by its S_3 normalizer; and A_4 cycles the three V_4 eigenlines, leaving all fifteen with exact stabilizer V_4 .

It remains to account for exact C_2 stabilizers. Each involution lies in two D_5 subgroups, two S_3 subgroups, and one V_4 . Of its thirteen fixed points, these contribute respectively 2, 2, and 3 points from the preceding rows. The remaining six points have exact stabilizer C_2 . There are fifteen involutions, so this gives 90 points, necessarily three orbits of length $60/2 = 30$. The total $6 + 10 + 12 + 15 + 90 = 133$ exhausts the plane. Dye’s Theorem 6(v), specialized at $q = 11 \equiv 1 \pmod{10}$, identifies the twelve-point C_5 -stabilizer orbit with $\mathcal{C}(\mathbb{F}_{11})$ [16, pp. 281–282; §3.1, p. 284]; the other three identifications are the transitive sets cited above. $\square$

The kernel certificate verifies this concrete sixty-element action and its seven orbits, not the abstract character argument.

Proposition 3.2 (The projective deep-hole syndrome locus is the conic). *The covering radius of C is three, and its projective deep-hole syndrome locus $\mathcal{D}_{\text{proj}}(C)$ — the set of directions $x \notin A$ on no secant of A — is exactly the twelve points of $\mathcal{C}$.*

Proof. The set $\mathcal{U}(A)$ is G -invariant. Dye’s Clebsch classification gives exactly ten Brianchon points. For completeness, here is the relevant double count. The fifteen edges contribute 150 off-arc incidences. The 45 pairs of disjoint edges meet once, so if n_i counts off-arc points lying on exactly i edges, then

$$n_1 + 2n_2 + 3n_3 = 150, \quad n_2 + 3n_3 = 45, \quad n_3 = 10.$$

Thus the edges cover $n_1 + n_2 + n_3 = 115$ of the 127 off-arc points, leaving $|\mathcal{U}(A)| = 12$. This is the $q = 11$ specialization of the chord-defect identity in Lemma 6.2. By definition $\mathcal{U}(A)$ is disjoint from the unique six-point orbit A . The orbit lengths in Proposition 3.1 therefore force this invariant twelve-set to be the unique twelve-point orbit $\mathcal{C}(\mathbb{F}_{11})$. By the dictionary

above this is exactly the projective deep-hole syndrome locus, and its nonemptiness gives covering radius three rather than two. Dye's edge criterion and the complete-exterior-set description of Blokhuis–Seress–Wilbrink independently supply the classical inclusion $\mathcal{C}(\mathbb{F}_{11}) \subseteq \mathcal{U}(A)$ [16, discussion preceding Theorem 6, p. 281] [2, pp. 143, 146]. $\square$

The companion arcs paper records this identification as one secant-index spectrum [7, Prop. 8.7]. We prove it here because it starts the rigidity theory; that theory, the gap, chirality, and the isolation of $q = 11$ are not in the companion paper.

Corollary 3.3 (Directions, cosets, leaders, and received words). *The projective deep-hole syndrome locus has 12 directions, exactly the points of $\mathcal{C}$. Above them lie $12(11 - 1) = 120$ nonzero maximum-distance syndromes, hence 120 deep-hole cosets. Each such coset has $\binom{6}{3} = 20$ minimum-weight leaders, for 2400 leaders in all, and contains $|C| = 11^3 = 1331$ received words. Consequently C has $120 \cdot 1331 = 159720$ received-word deep holes.*

Proof. Immediate from Proposition 3.2, the leader-count clause of the DMP dictionary, and the fact that the fiber of $\mathbb{F}_{11}^3 \setminus \{0\} \rightarrow \text{PG}(2, 11)$ over each direction consists of ten nonzero syndromes, hence ten distinct cosets. Every coset has $|C|$ words. $\square$

Proposition 3.4 (One orbit on received-word deep holes). *The monomial automorphism group of C is transitive on the 120 deep-hole cosets. Let Γ be the group generated by these automorphisms and by translations $v \mapsto v + c$ for $c \in C$. Then*

$$|\Gamma| = |C| |\text{MAut}(C)| = 1331 \cdot 600 = 798600,$$

and Γ is transitive on all 159720 received-word deep holes. The stabilizer of each such word is cyclic of order five.

Proof. The projective stabilizer A_5 is transitive on the twelve points of $\mathcal{C}$. Indeed, its six Sylow 5-subgroups form one conjugacy class, each fixes the two endpoints of one of the six axis chords, and the involution in its dihedral normalizer interchanges those endpoints. These twelve endpoints exhaust $\mathcal{C}(\mathbb{F}_{11})$. Every such projectivity lifts to a monomial automorphism of the parity-check columns, so the monomial group is transitive on the twelve syndrome rays above $\mathcal{C}$. Its central scalar subgroup $\mathbb{F}_{11}^\times$ acts transitively on the ten nonzero vectors of each ray. Hence the monomial group is transitive on all $12 \cdot 10 = 120$ maximum-distance syndromes and therefore on the corresponding cosets. Given deep holes v and w , choose $m \in \text{MAut}(C)$ with $\sigma(mv) = \sigma(w)$. Then $c = w - mv$ has zero syndrome, so $c \in C$, and the codeword translation by c sends mv to w . Both kinds of generator are Hamming isometries preserving C . The monomial group normalizes the translation subgroup, since $mt_c m^{-1} = t_{m(c)}$ with $m(c) \in C$. Thus $\Gamma = C \rtimes \text{MAut}(C)$; the two subgroups have trivial intersection and the displayed product order follows. Orbit–stabilizer now gives stabilizer order $798600/159720 = 5$, and every group of prime order is cyclic. $\square$

The coset-leader-weight distribution of C is $(1, 60, 1150, 120)$ (cosets of weight 0, 1, 2, 3, summing to $11^3 = 1331$); in particular the last entry counts deep-hole cosets, not received words. Its codeword weight enumerator is $(1, 0, 0, 0, 150, 420, 760)$, forced by the MDS parameters; we record it as a sanity check, not a novelty.

The special geometry also makes distance diagnosis and nearest-word structure unusually explicit. Write a syndrome as $s = (s_0, s_1, s_2)$ and put $Q(s) = s_0 s_2 - s_1^2$.

Proposition 3.5 (Complete syndrome oracle and ambiguity enumerator). *For $v \in \mathbb{F}_{11}^6$ with syndrome $s = Hv^\top$,*

$$d(v, C) = \begin{cases} 0, & s = 0, \\ 1, & [s] \in A, \\ 3, & s \neq 0 \text{ and } Q(s) = 0, \\ 2, & \text{otherwise.} \end{cases}$$

Among the 1330 nonzero cosets, the numbers having respectively one, two, three, and twenty nearest codewords are

$$960, \quad 150, \quad 100, \quad 120.$$

For a distance-two syndrome the number of nearest codewords is its secant index. Every distance-three syndrome has exactly one leader on each of the twenty three-element coordinate supports.

Proof. The DMP dictionary identifies weight-one directions with the six columns, weight at most two with the secant union, and weight three with its complement. Proposition 3.2 identifies the last set with $Q = 0$, proving the oracle. In the notation of its proof, the chord-intersection equations give $(n_0, n_1, n_2, n_3) = (12, 90, 15, 10)$, where n_i counts projective directions of secant index i and n_0 is the uncovered locus. Every direction has ten nonzero affine syndromes. Thus the distance-two cosets contribute 900 cases of multiplicity one, 150 of multiplicity two, and 100 of multiplicity three; the sixty weight-one cosets add sixty more cases of multiplicity one. Leaders are in bijection with nearest codewords by translation within the coset. Finally, the three columns on any support are independent. For an uncovered syndrome their unique coefficients are all nonzero, since a zero coefficient would put the syndrome on a secant. Hence all twenty supports, and no others, give its weight-three leaders. $\square$

For this fixed code the proposition is a constant-size post-syndrome test, not an asymptotic claim. The replay and kernel certificate check the oracle, support theorem, and ambiguity counts.

3.1 Certified structural facts

The following consequences of the MDS parameters and classical geometry anchor the code's symmetry and design structure.

- **Automorphism groups.** The projective stabilizer of the six column rays is A_5 of order 60, acting faithfully and 2-transitively on the six coordinate supports in its exotic degree-six action. It is the support-permutation quotient of the monomial automorphism group, which fits into the central exact sequence

$$1 \longrightarrow \mathbb{F}_{11}^\times \longrightarrow \text{MAut}(C) \longrightarrow A_5 \longrightarrow 1.$$

The kernel consists of the ten global coordinate scalars. The sequence splits: since $\gcd(3, 10) = 1$, every class in $\text{PGL}(3, 11)$ has a unique determinant-one representative, and these representatives multiply; their associated monomial lifts give an A_5 complement. Thus $\text{MAut}(C) \cong C_{10} \times A_5$ has order 600. By contrast, for the displayed choice of column representatives the subgroup of pure coordinate permutations is trivial. An exact computation confirms these group orders and the transitive action on the 120 maximum-distance syndromes above C .

- **Design view.** The MDS parameters make C an orthogonal array $\text{OA}(11^3, 6, 11, 3)$ of strength three.
- **Five self-polar triangles.** Edge’s finite-plane construction [5, §§30.2–31] and Dye’s general-field synthetic theory [16, Theorem 2, pp. 277–278] exhibit five triangles on the six vertices, self-polar for $\mathcal{C}$. Their vertex-pair matchings partition the fifteen pairs into a synthemetic total; Section 5 turns its ten triangle pairs into the Brianchon–support dictionary.

4 The rigidity theorem

The content of Proposition 3.2 is, on its own, a short synthesis for one known arc, independently certified by finite computation. The following theorem is the paper’s main claim: among *all* six-arcs of $\text{PG}(2, 11)$, the property “the projective deep-hole syndrome locus lies on a conic” picks out the Clebsch hexagon uniquely, and does so in a way that recovers its projective stabilizer from the coding condition rather than assuming it.

Remark 4.1 (Two conditions, one word apart). The condition studied here is that $\mathcal{U}(A)$ — the projective uncovered, or projective deep-hole syndrome, locus — lies on a conic. It is *not* the familiar condition that the arc A itself lies on a conic, which is the standing hypothesis throughout the arc literature and which the Clebsch hexagon famously fails. The two are logically unrelated: they constrain different point sets, and the arcs satisfying them are disjoint. Indeed every six-arc lying *on* a conic in $\text{PG}(2, 11)$ has $|\mathcal{U}(A)| \in \{18, 19, 20, 22\}$, so by (iii) none of them comes anywhere near the condition of the theorem.

Census framing. The projective-equivalence classes of k -arcs in $\text{PG}(2, 11)$ were classified by Sadeh [24]; see also Hirschfeld [19, Ch. 14]. Over the six-arc classes of that census we compute, for each representative A , the projective uncovered locus $\mathcal{U}(A)$ — the points off A and off all fifteen of its secants, equivalently the points extending A to a projective seven-arc — and read $\mathcal{U}(A)$ as the projective covering-radius-three syndrome data of the associated $[6, 3, 4]_{11}$ code. *We claim no priority on the six-arc census, nor on the raw extension-count data $|\mathcal{U}(A)|$ that an extension-based classification entails:* the size of $\mathcal{U}(A)$ is a standard projective invariant of any arc, and the census itself is Sadeh’s. What we contribute is the geometric and coding-theoretic meaning of $\mathcal{U}(A)$: that its containment in a conic singles out the Clebsch hexagon and forces A_5 , that this containment is rigid rather than merely typical, and the resulting gap, decoding, and support structure. The exact conic locus for the singled-out class is the setup supplied by Proposition 3.2, not a separate novelty claim here.

We first isolate the geometric input that removes degenerate conics from the computer-assisted part of the argument.

For an external line, the secant intersections form the focus set studied in the line-covering literature [23]. Equality at five covered points would make the six-arc hyperfocused. The general internal-nucleus theorem of Bichara and Korchmáros excludes nontrivial hyperfocused arcs in odd order [3]; the six-point case also follows from Drake and Keating’s classification of hyperovals in five-nets [6, Theorem 3.10]. We retain the short direct argument because it records the chord, one-vertex, and external-line cases together in exactly the form needed below.

Lemma 4.2 (Six-arc line bound). *Let A be a six-arc in $\text{PG}(2, q)$, where q is odd. Every line contains at most $q - 5$ points of $\mathcal{U}(A)$.*

Proof. A chord contains no point of $\mathcal{U}(A)$. If a non-chord line ℓ contains one vertex, the ten chords on the other five vertices meet ℓ in at least five distinct points: their intersections give a proper edge-colouring of K_5 , whose chromatic index is five. Together with the vertex, at least six of the $q + 1$ points of ℓ are therefore outside $\mathcal{U}(A)$.

It remains to consider $\ell \cap A = \emptyset$. The fifteen chords meet ℓ in covered points, and at most three chords pass through any one of them, because their endpoint pairs are disjoint. Hence there are at least five covered points. Equality would give a proper five-edge-colouring of K_6 , necessarily a one-factorization. After relabelling, three factors are

$$01|23|45, \quad 02|14|35, \quad 03|15|24.$$

Send ℓ to infinity and normalize the first two directions to horizontal and vertical. Put $p_0 = (0, 0)$, $p_1 = (x, 0)$ and $p_2 = (0, y)$, where $xy \neq 0$. Writing $p_3 = (a, y)$, the first two factors force $p_4 = (x, b)$ and $p_5 = (a, b)$. Parallelism of 03 with 15 and with 24 gives two determinant equations whose difference is $2xy = 0$, impossible for odd q . Thus a disjoint line has at least six covered points, and in all cases at most $(q + 1) - 6 = q - 5$ points of $\mathcal{U}(A)$ remain. $\square$

Remark 4.3 (Frame-normalized census). Any ordered four points of a six-arc form a projective frame, and a unique projectivity sends them to $(1, 0, 0), (0, 1, 0), (0, 0, 1), (1, 1, 1)$. We enumerate the remaining unordered pair subject to the arc condition, obtaining 1548 normalized sets. The lexicographically least of a set's $6 \cdot 5 \cdot 4 \cdot 3 = 360$ ordered-frame normalizations is a complete projective-class key; there are fifteen keys. A stabilizer acts freely on ordered frames, so a class with stabilizer G contributes $360/|G|$ normalized sets. The Clebsch class contributes six.

Theorem 4.4 (Rigidity theorem). *For a six-arc $A \subset \text{PG}(2, 11)$, the following are equivalent:*

- (i) *the projective deep-hole syndrome locus $\mathcal{U}(A)$ lies on some conic (possibly degenerate);*
- (ii) *$\mathcal{U}(A)$ is exactly the set of $\mathbb{F}_{11}$ -points of a nonsingular conic;*
- (iii) *$|\mathcal{U}(A)| \leq 15$ (in fact, $|\mathcal{U}(A)| = 12$);*
- (iv) *A is $\text{PGL}(3, 11)$ -equivalent to the Clebsch hexagon;*
- (v) *the stabilizer of A in $\text{PGL}(3, 11)$ contains A_5 .*

Proof. Suppose first that $\mathcal{U}(A)$ lies on a conic Q . If Q is nonsingular then $|\mathcal{U}(A)| \leq q + 1 = 12$. If Q is degenerate, its rational points lie on one or two rational lines when it splits; in the nonsplit rank-two case its only rational point is the singular point. Thus Lemma 4.2 again gives $|\mathcal{U}(A)| \leq 12$. On the other hand, the chord-defect identity of Lemma 6.2 says

$$|\mathcal{U}(A)| = 22 - c(A),$$

where $c(A)$ is the number of nonvertex triple-edge concurrences. In Dye's terminology these are precisely the Brianchon points of the hexagon A . Over a field of characteristic different from two, Dye proves that a hexagon has at most ten Brianchon points and that the equality cases form a single $\text{PGL}_3(K)$ -orbit [16, §2.2–2.3, Theorem 1, pp. 275–276]. Hence $c(A) \leq 10$, so $|\mathcal{U}(A)| \geq 12$; equality holds throughout, $c(A) = 10$, and A is a Clebsch hexagon. This proves (i) $\Rightarrow$ (iv). Proposition 3.2 and projective equivariance give (iv) $\Rightarrow$ (ii) $\Rightarrow$ (i). Dye's stabilizer theorem gives (iv) $\Rightarrow$ (v), and Storme and Van Maldeghem's finite-plane classification gives the converse.

It remains only to connect the numerical condition (iii). The complete census of Remark 4.3 yields

$$|\mathcal{U}(A)| \in \{12, 16, 18, 19, 20, 21, 22\}.$$

The corresponding representative multiplicities are

$$(6, 30, 150, 300, 630, 360, 72).$$

Only the six Clebsch representatives attain 12; every other class has at least 16 extension points. Thus (iii) $\Leftrightarrow$ (iv), completing the equivalence. The exhaustive calculation is therefore used for the size-gap clause, not for the conic- containment implication. $\square$

The standalone artifact `check_rigidity_degenerate_conic.py` implements the exhaustive computation just described, reproduces the displayed histogram, and returns exactly the six normalized Clebsch-class representatives whose extension loci lie on conics; every containing conic is nonsingular.

The formal development exposes the classical trust seam in this proof explicitly. Dye’s Theorem 1 is its only specially axiomatized mathematical source input, represented by two statements in `lean/RelativeConicArcs/Q11DyeAxioms.lean`: the ten-point Brianchon bound and the classification of equality. The consequences used above are derived in `Q11DyeConsequences.lean`; the chord-defect identity and the line bound are proved independently of those axioms. At $q = 11$ the census also checks the imported conclusion by a separate finite route: its minimum $|\mathcal{U}(A)| = 12 = 22 - 10$ gives $\max c(A) = 10$, and its six normalized minimizers form the single Clebsch class.

The same census supports a strengthening in which a cubic, rather than a conic, is allowed.

Proposition 4.5 (Low-degree algebraic rigidity). *For a six-arc $A \subset \text{PG}(2, 11)$, the uncovered locus $\mathcal{U}(A)$ is contained in the zero set of a nonzero homogeneous form of degree at most three if and only if A is projectively equivalent to the Clebsch hexagon. In the Clebsch case the least such degree is two: the quadratic vanishing space is spanned by the defining nonsingular conic equation Q , and the cubic vanishing space is*

$$Q \cdot \mathbb{F}_{11}[X, Y, Z]_1.$$

Computer-assisted proof. For each of the fifteen projective classes in the six-arc census, evaluate the ten cubic monomials on $\mathcal{U}(A)$. The resulting evaluation map has rank ten for every non-Clebsch class and rank seven for the Clebsch class. Thus no nonzero cubic vanishes on a non-Clebsch uncovered locus. A containing form of smaller positive degree could be multiplied by a nonzero monomial to produce such a cubic, so smaller positive degrees are excluded as well; no nonzero constant vanishes on the nonempty locus. For the Clebsch class, the quadratic kernel is generated by Q , and multiplication by linear forms gives the full three-dimensional cubic kernel. The exact evaluation replay checks all ranks and kernel identities. $\square$

Corollary 4.6 (Monomial characterization). *Up to monomial equivalence, C is the unique linear $[6, 3, 4]_{11}$ code of covering radius three whose projective deep-hole syndrome locus is contained in a plane curve of degree at most three. In particular, A_5 is recovered from this coding condition, not assumed as a hypothesis.*

Proof. The columns of a parity-check matrix of such a code form a six-arc A , and covering radius three identifies its projective deep-hole syndrome locus with $\mathcal{U}(A)$. Proposition 4.5

makes A projectively equivalent to the Clebsch hexagon. A projectivity between two unordered parity-check column sets lifts, after choosing column representatives, to a row operation together with a coordinate permutation and nonzero coordinate scalings; these are exactly a monomial code equivalence. The converse is immediate, and the stabilizer statement is clause (v) of the theorem. $\square$

Remark 4.7 (Sharpness of the degree threshold). Degree three is best possible. Apart from the Clebsch conic, the exhaustive search through degree five finds precisely two projective classes whose uncovered locus is the full set of $\mathbb{F}_{11}$ -rational zeros of a form: one 18-point locus is a smooth absolutely irreducible quartic of genus three, and one 19-point locus is an absolutely irreducible quintic with one ordinary nonsplit node, whose normalization has genus five. The fail-closed Singular replay also finds sixteen exact sextics in the first nonzero kernel of one 22-point class, including a smooth absolutely irreducible witness. We make no degree-six classification: the larger degree-six kernels of the other classes were not exhausted.

Remark 4.8 (Which implications are new). The equivalence (iv) $\Leftrightarrow$ (v) is imported rather than new here. Dye proves projective uniqueness of Clebsch hexagons and computes their stabilizer as A_5 in characteristic eleven [16, Theorems 1 and 3, pp. 275–278], giving (iv) $\Rightarrow$ (v). Storme and Van Maldeghem classify the primitive A_5 -fixed arcs and prove projective uniqueness of their six-point orbit [27, Prop. 12], giving (v) $\Rightarrow$ (iv). The content of the theorem is the implication (i) $\Rightarrow$ (iv), and with it the equivalence of the projective syndrome conditions (i)–(iii) with the classical characterizations (iv)–(v). It is that bridge — from a covering-radius property of the code to a projective classification — that is the claim, and the corollary above is its consequence.

The rigidity theorem has three quantitative refinements: a global gap in the extension-count census, a projectively invariant distance from the nearest conic, and a local gap under single-vertex replacement. The first is a statement of the standard extremal shape — there is no object strictly between the extremal value and the next, and the objects attaining the extremal value are classified — familiar from the minimum-side gap results for arcs and blocking sets, such as Blokhuis–Bruen’s [14] theorem on the sizes realized just above the minimum. Here the invariant is the projective deep-hole-direction count $|\mathcal{U}(A)|$.

For the second, let $\mathfrak{C}$ be the set of nonsingular conics in $\text{PG}(2, 11)$ and define the *nearest-conic discrepancy*

$$\delta(A) = \min_{\mathcal{Q} \in \mathfrak{C}} |\mathcal{U}(A) \triangle \mathcal{Q}(\mathbb{F}_{11})|.$$

This is a PGL -invariant function, not a metric on arcs: equivariance gives $\mathcal{U}(gA) = g\mathcal{U}(A)$, while PGL permutes $\mathfrak{C}$.

For the local statement, let $\mathcal{G}_1$ be the graph whose vertices are embedded six-arcs in $\text{PG}(2, 11)$, with two vertices adjacent when their point sets have symmetric difference two. Fix the displayed Clebsch arc A and its standard conic $\mathcal{C}$, and put

$$d_{\mathcal{C}}(B) = |\mathcal{U}(B) \triangle \mathcal{C}|$$

for every embedded six-arc B ; clause (c) restricts this discrepancy to the neighbors of A .

Theorem 4.9 (Gap theorem). *The Clebsch configuration is rigid rather than merely stable.*

- (a) (Census gap.) *There is no six-arc $A \subset \text{PG}(2, 11)$ with $12 < |\mathcal{U}(A)| < 16$. Furthermore, every six-arc attaining $|\mathcal{U}(A)| = 12$ is $\text{PGL}(3, 11)$ -equivalent to the Clebsch hexagon, and every six-arc with $|\mathcal{U}(A)| \geq 16$ has $\mathcal{U}(A)$ contained in no conic.*

(b) (Global nearest-conic gap.) *On the fifteen $\mathrm{PGL}(3, 11)$ -classes of six-arcs, the exact class-counted histogram of δ is*

$$\{0:1, 12:2, 13:1, 14:3, 15:1, 16:4, 17:2, 18:1\}.$$

Its unique zero is the Clebsch class. Thus every non-Clebsch class has $\delta(A) \geq 12$, and the bound is sharp.

(c) (Perturbation gap.) *Each deleted vertex of A has exactly 42 legal replacements, giving 252 distinct neighbors B . Their exact fixed-conic discrepancy histogram is*

$$\{18:30, 19:60, 20:90, 22:42, 24:30\}.$$

The corresponding histogram of $|\mathcal{U}(B) \cap \mathcal{C}|$ is $\{4:60, 6:132, 7:60\}$, so at most seven conic directions remain; no neighbor has $\mathcal{U}(B)$ contained in any conic, degenerate or otherwise. Under $\mathrm{Stab}(A) \cong A_5$, the neighbors split into eight orbits whose pairs (orbit size, fixed-conic discrepancy) are

$$(30, 18), (60, 19), (30, 20), (30, 20), (30, 20), (12, 22), (30, 22), (30, 24).$$

Thus every vertex adjacent to A in $\mathcal{G}_1$ has $d_{\mathcal{C}}(B) \geq 18$.

Remark 4.10 (Global versus fixed-conic discrepancy). The closing statement is local to the fixed pair $(A, \mathcal{C})$ and is not a global nearest-arc theorem. The conic-preserving projectivity $g : (X, Y, Z) \mapsto (4X, 2Y, Z)$ sends A to a distinct embedded six-arc, while equivariance gives $\mathcal{U}(gA) = g\mathcal{U}(A) = \mathcal{C}$. Hence $d_{\mathcal{C}}(gA) = 0$, although $gA \neq A$. This is exactly why the global statement minimizes over conics and passes to projective classes: gA remains in the Clebsch class and has $\delta(gA) = 0$. The lower bound eighteen is specifically for the neighbors of the fixed embedded A in $\mathcal{G}_1$; the global non-Clebsch lower bound is the separate value twelve.

Remark 4.11 (Two spectra, one word apart). Clause (a) is a gap in the spectrum of the *projective deep-hole- direction count* $|\mathcal{U}(A)|$ at fixed arc size $k = 6$. It is not a statement about the much-studied spectrum of the *sizes* k for which a complete k -arc exists, which at $q = 11$ was determined by Faina, Marcugini, Milani and Pambianco [15]: there the invariant is k itself and the relevant constants are $t(2, 11) = 7$ and $m''(2, 11) = 10$. The two meet only at the boundary $\mathcal{U}(A) = \emptyset$, where an arc is complete; that $t(2, 11) = 7$ exceeds six is exactly why no six-arc of $\mathrm{PG}(2, 11)$ is complete, and hence why the histogram above has no zero bin.

Computer-assisted proof. The first clause restates the exhaustive classification of Theorem 4.4. For the second, `check_global_conic_gap.py` normalizes all 360 ordered four-frames of each of the 1548 representatives to obtain exactly fifteen projective-class keys. Independently it enumerates all 177156 projective ternary quadrics, retains exactly 160930 nonsingular conics, and computes the minimum symmetric difference for each class. It asserts the displayed class histogram, a direct-XOR/intersection-formula cross-check, and the unique zero class.

For the third clause, the local checker enumerates the 133 projective points, deletes each of the six vertices in turn, and tests every replacement outside A for the six-arc condition. It asserts the six per-vertex counts, distinctness of the 252 resulting arcs, both displayed histograms, and conic noncontainment by exact quadratic-monomial rank. It then uses the sixty exact projectivities in the stabilizer to certify the eight closed orbits and recover both histograms from their rows. It also verifies the projectivity in the preceding remark and directly recomputes $\mathcal{U}(gA) = \mathcal{C}$. Its source is `check_perturbation_gap.py`. $\square$

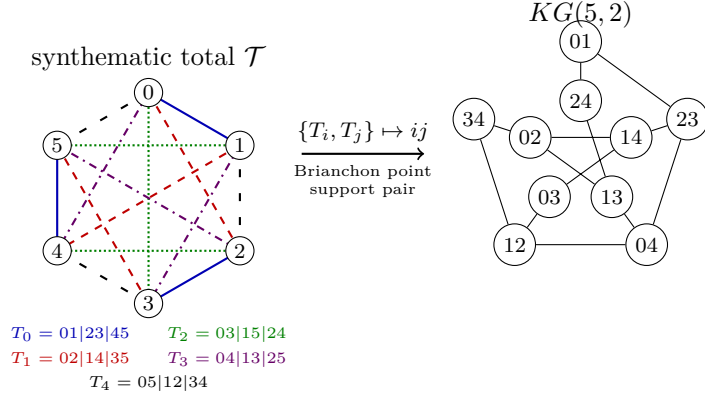


Figure 1: The synthemetic–Petersen dictionary. The five self-polar triangles give the displayed perfect matchings. A Petersen vertex ij represents the Brianchon point and complementary support pair attached to $\{T_i, T_j\}$; two vertices are adjacent when the index pairs are disjoint.

Remark 4.12 (Why these facts are not already recorded). The locus $\mathcal{U}(A)$ is the classical extension locus; see Ball and Lavrauw [10, Thms. 39–41]. Sadeh’s census uses six-arcs to construct cubic surfaces and has no reason to compare their extension loci with conics. Conversely, the tangent-envelope theory that studies extension loci is effective for much larger arcs. The new point here is the coding interpretation of $\mathcal{U}(A)$ and the rigidity detected at six points.

5 The chirality bipartition

The 20 three-supports form two complementary A_5 -orbits of size ten. Their ten complementary pairs carry the Petersen graph described in Figure 1; the next proposition identifies its vertices geometrically.

Proposition 5.1 (Brianchon–support dictionary). *Let $\mathcal{T}$ be the five self-polar triangles of Edge and Dye [5, 16], regarded as the five perfect matchings in their synthemetic total, and let $\mathcal{B}(A)$ be the set of ten Brianchon points of the Clebsch hexagon. For distinct $T, T' \in \mathcal{T}$, the union $T \cup T'$ is an alternating six-cycle. Let $M(T, T')$ be the perfect matching joining opposite vertices of this cycle, and let $\beta(T, T') = \{S, S^c\}$ be its unordered bipartition.*

The three chords indexed by $M(T, T')$ concur at a point $b(T, T') \in \mathcal{B}(A)$. The maps

$$\binom{\mathcal{T}}{2} \longrightarrow \mathcal{B}(A), \quad \{T, T'\} \longmapsto b(T, T'),$$

and

$$\binom{\mathcal{T}}{2} \longrightarrow \{\{S, S^c\} : |S| = 3\}, \quad \{T, T'\} \longmapsto \beta(T, T'),$$

are A_5 -equivariant bijections. They identify the ten Brianchon points with the ten complementary support pairs. Under this identification, Petersen adjacency is disjointness of the corresponding two-element subsets of $\mathcal{T}$.

Proof. The opposite-vertex matching is disjoint from T and T' and contains one edge from each of the other three members of the synthemetic total. Hence $\{T, T'\}$ is recovered as the two members of $\mathcal{T}$ disjoint from $M(T, T')$, so the ten triangle pairs give exactly the ten

perfect matchings outside the total. Edge’s Clebsch construction says that the three chords of each such matching concur, at the ten distinct Brianchon points [5, §§19–20,30]. The bipartition bijection and Petersen assertion are the alternating-cycle construction above. All concurrences, the ten multiplicity-three and fifteen multiplicity-one intersections of disjoint chords, and the equivariance and support-map compatibility follow from these two equivariant bijections. $\square$

Corollary 5.2 (The decoder reconstructs the Brianchon configuration). *The ten projective directions having three nearest weight-two errors are exactly the ten Brianchon points. At each such direction the three error supports are pairwise disjoint and hence form a perfect matching of the six coordinates. The ten matchings thereby recovered are precisely the ten perfect matchings outside the five-matching synthematic total.*

Proof. The three weight-two representations of a secant-index-three direction use the three chords through it. Proposition 5.1 identifies the ten triple concurrences with the ten Brianchon points and their chords with exactly the ten matchings complementary to the total. $\square$

Two classical constructions are adjacent but different. Edge’s two systems of eleven hexagons over a conic concern classes of hexagons, not support orbits [5, §§29,32]. The Petersen graph in the $q = 31$ exterior set of Blokhuis–Seress–Wilbrink records secants of a ten-set, not the present $q = 11$ chirality [2, p. 146].

Proposition 5.3 (Invariant support chirality). *The twenty three-element coordinate supports split into two complementary A_5 -orbits $\mathcal{O}_+$ and $\mathcal{O}_-$ of size ten. For every maximum-distance syndrome s , each support determines exactly one weight-three leader of the coset $H^{-1}(s)$; hence that coset has ten leaders supported in each class. Globally, the 2400 minimum-weight leaders split as $1200 + 1200$.*

Every monomial automorphism of C preserves this unordered bipartition. The other coset in the exotic normalizer $N_{S_6}(A_5) \cong S_5$ exchanges the two support classes, but its sixty elements are not automorphisms of the displayed code. Thus the intrinsic structure is a two-class decomposition, or an unbased $\mathbb{Z}/2$ -torsor, rather than a canonically oriented $\mathbb{Z}/2$ -valued label. By Proposition 5.1, the ten complementary support pairs are canonically both the two-subsets of the five self-polar triangles and the ten Brianchon points.

Proof. The exotic degree-six A_5 has two complementary ten-orbits on the 20 three-supports. It preserves a unique synthematic total, also preserved by its full S_5 normalizer. The alternating-cycle construction of Figure 1 is therefore equivariant, and the outside normalizer coset exchanges the two support orbits. Fix a maximum-distance syndrome s and a three-support S . The three columns indexed by S are independent because A is an arc, so they give a unique coefficient vector representing s . None of its three coefficients can vanish: otherwise $[s]$ would lie on a secant of A , contrary to maximum distance. The resulting error vector therefore has weight three and is the unique leader with support S . Finally, every monomial automorphism induces a support permutation in A_5 , by the exact sequence in Section 3.1, and so preserves each support orbit. The outside normalizer coset swaps the two orbits, but the pure-permutation audit there and the monomial support-image audit show that it does not preserve C , even after coordinate rescaling. $\square$

The finite replay checks the full dictionary, normalizer, leaders, and monomial action.

The bipartition is therefore intrinsic to the code’s covering-radius data, but its two sides have no preferred signs without an additional orientation.

There is a finer equivariance distinction. The stabilizer in $\text{MAut}(C)$ of one *affine* deep-hole syndrome has order five and four orbits of size five on its twenty leaders. Transporting

any one orbit gives a monomial-equivariant set-valued decoder of constant list size five on the 120 deep-hole syndromes, and every nonempty equivariant decoder on this syndrome orbit has list size at least five. Thus chirality is not the minimum or coarsest arbitrary equivariant selection: its two size-ten lists are the proper nonempty selections determined solely by the support, each a union of two of the size-five orbits. Exact transport checks all 288000 equivariance cases without choosing an orientation of the two chirality classes.

6 Why $q = 11$

The conic-filling syndrome phenomenon of Sections 3–4 is specific to $q = 11$, even without a symmetry hypothesis on the arc. A universal chord-multiplicity identity makes this a four-field question, and the only nontrivial residue is a classical exact-distance problem in the Sylvester graph.

Lemma 6.1 (Secant-covering bound). *Let A be a six-arc of $\text{PG}(2, q)$ disjoint from a non-singular conic $\mathcal{C}$, and suppose A is complete outside $\mathcal{C}$: every point of $\text{PG}(2, q)$ lying off A and off $\mathcal{C}$ lies on some secant of A . Then*

$$15(q - 1) \geq q^2 - 6,$$

equivalently $q^2 - 15q + 9 \leq 0$, so $q \leq 14$.

Proof. Since A has six points and $\mathcal{C}$ has $q + 1$ points, and A is disjoint from $\mathcal{C}$,

$$|\text{PG}(2, q) \setminus (A \cup \mathcal{C})| = (q^2 + q + 1) - 6 - (q + 1) = q^2 - 6.$$

Each of the $\binom{6}{2} = 15$ secants of A is a line of $\text{PG}(2, q)$, hence has $q + 1$ points, exactly two of which lie in A ; so each secant covers at most $q - 1$ points of $\text{PG}(2, q) \setminus A$, and in particular at most $q - 1$ points of $\text{PG}(2, q) \setminus (A \cup \mathcal{C})$. Completeness outside $\mathcal{C}$ says the fifteen secants together cover all $q^2 - 6$ points of this set, so

$$q^2 - 6 \leq 15(q - 1).$$

(No accounting for overlap between secants is needed here: the bound only uses that fifteen sets of size at most $q - 1$ cover a set of size $q^2 - 6$, which requires the sum of their sizes to be at least $q^2 - 6$ regardless of how much they overlap; overlap can only make the true covered count smaller than the sum, never larger, so the inequality direction needed for an upper bound on q is exactly the one overlap cannot violate.) Rearranging, $q^2 - 15q + 9 \leq 0$, whose real roots are $\frac{15 \pm \sqrt{189}}{2} \approx 0.62, 14.38$; since q is a positive integer, $q \leq 14$. $\square$

Lemma 6.2 (Chord-defect identity). *Let A be a six-arc in $\text{PG}(2, q)$, and let $c(A)$ be the number of points off A incident with three of its fifteen chords. Then*

$$|\mathcal{U}(A)| = q^2 - 14q + 55 - c(A), \quad 0 \leq c(A) \leq 15.$$

Consequently, if $|\mathcal{U}(A)| = q + 1$, then

$$c(A) = (q - 6)(q - 9).$$

Proof. At a point off A , concurrent chords have pairwise disjoint endpoint pairs, so at most three chords occur. Let n_i count the off-arc points on exactly i chords. Counting chord–point incidences and pairs of chords with disjoint endpoints gives

$$n_1 + 2n_2 + 3n_3 = 15(q - 1), \quad n_2 + 3n_3 = \binom{15}{2} - 6\binom{5}{2} = 45.$$

Writing $c(A) = n_3$, the number of covered points off A is $n_1 + n_2 + n_3 = 15q - 60 + c(A)$. Subtracting this from the $q^2 + q - 5$ points off A proves the identity. Each triple concurrence uses one of the fifteen perfect matchings of the six endpoints, and a fixed matching can be concurrent at only one point; hence $c(A) \leq 15$. The last display follows on setting $|\mathcal{U}(A)| = q + 1$. $\square$

Theorem 6.3 (Uniqueness of $q = 11$). *Let q be a prime power. If a six-arc $A \subset \text{PG}(2, q)$ satisfies $\mathcal{U}(A) = \mathcal{C}(\mathbb{F}_q)$ for some nonsingular conic $\mathcal{C}$, then $q = 11$. Moreover, at $q = 11$ every such A is projectively equivalent to the Clebsch hexagon.*

Proof. Lemma 6.2 gives $c(A) = (q - 6)(q - 9)$ with $0 \leq c(A) \leq 15$. The prime-power condition therefore leaves only $q \in \{4, 5, 9, 11\}$: the values at 7 and 8 are negative, while every integer $q \geq 12$ gives $c(A) \geq 18$.

At $q = 4$, every six-arc is a hyperoval. Each point off it lies on three secants (count the six hyperoval points on the five lines through the point), so $\mathcal{U}(A)$ is empty. At $q = 5$, every six-arc is an oval and hence a nonsingular conic; the standard internal/external point counts show that every off-conic point lies on a secant, so again $\mathcal{U}(A)$ is empty.

It remains to exclude $q = 9$. Relative to the hypothetical conic $\mathcal{C}$, every chord of A is passant. A point off $\mathcal{C}$ lies on five passants if it is internal and four if it is external; hence the five chords through each vertex force all six vertices of A to be internal. Let H be the graph on the 36 internal points, with $P \sim Q$ when $Q \in P^\perp$ under the conic polarity. Standard conic counts give the intersection array

$$\{5, 4, 2; 1, 1, 4\},$$

so H is the Sylvester graph [12, §13.1.2]; see also the combinatorial uniqueness proof in [13, Thm. 6]. For distinct internal points, the unique possible common neighbor is $P^\perp \cap Q^\perp = (PQ)^\perp$; consequently PQ is passant exactly when P, Q are at distance two in H . Thus A would be a six-clique in the exact-distance-two graph of H . But [11, Table 1] gives $\text{eq}_2(H) = 5$, a contradiction.

Therefore $q = 11$. The final assertion follows from Theorem 4.4, which puts every matching $q = 11$ representative in the single Clebsch orbit. $\square$

The from-scratch certificate independently constructs $\mathbb{F}_9$, $\text{PG}(2, 9)$, the polarity graph and its full intersection array, verifies passant join if and only if distance two for all internal-point pairs, and computes the exact clique number 5. A separate exact census follows.

For independent verification, every six-arc contains an ordered projective four-frame, which may be normalized to $(1, 0, 0), (0, 1, 0), (0, 0, 1), (1, 1, 1)$. Enumerating the remaining unordered pair with exact field arithmetic gives the following census; exponents in the third column record multiplicities.

q	representatives	histogram of $ \mathcal{U}(A) $	conic matches
2	0	—	0
3	0	—	0
4	1	0^1	0
5	3	0^3	0
7	70	$0^{40}, 2^{30}$	0
8	195	$0^{45}, 4^{150}$	0
9	441	$0^6, 4^{15}, 6^{120}, 7^{120}, 8^{180}$	0
11	1548	$12^6, 16^{30}, 18^{150}, 19^{300}, 20^{630}, 21^{360}, 22^{72}$	6
13	4015	$36^{85}, 38^{210}, 39^{480}, 40^{1080}, 41^{1800}, 42^{360}$	0

The enumeration uses exact field arithmetic, including explicit asserted tables for $\mathbb{F}_4, \mathbb{F}_8$, and $\mathbb{F}_9$; its conic test is also exercised on nonsingular and singular forms in characteristic two. It is reproduced by `check_small_q_uniqueness.py` and agrees with the conceptual proof above.

Proposition 6.4 (The Clebsch-family uncovered formula). *Let H be a Clebsch hexagon over $\mathbb{F}_q$. Then*

$$|\mathcal{U}(H)| = q^2 - 14q + 45.$$

If $q \equiv 3 \pmod{4}$ and $\mathcal{C}$ is Dye's associated conic, then $\mathcal{C}(\mathbb{F}_q) \subseteq \mathcal{U}(H)$ and

$$|\mathcal{U}(H) \setminus \mathcal{C}(\mathbb{F}_q)| = (q - 4)(q - 11).$$

In particular, among Clebsch hexagons in this congruence class, the associated conic is the complete projective deep-hole syndrome locus if and only if $q = 11$.

Proof. Dye defines a Clebsch hexagon by the occurrence of exactly ten Brianchon points, and his Theorem 1 classifies their existence and projective orbit [16, p. 271; Theorem 1, pp. 275–276]. These points are precisely the off-vertex triple-chord concurrences counted by $c(H)$. Thus $c(H) = 10$, and Lemma 6.2 gives

$$|\mathcal{U}(H)| = q^2 - 14q + 55 - 10 = q^2 - 14q + 45.$$

Dye also shows that the edges of H are non-secants of the associated conic when $q \equiv 3 \pmod{4}$ [16, discussion preceding Theorem 6, pp. 281–282]. Hence every rational point of $\mathcal{C}$ is uncovered. Subtracting $|\mathcal{C}(\mathbb{F}_q)| = q + 1$ from the first formula gives

$$q^2 - 15q + 44 = (q - 4)(q - 11).$$

The factor vanishes only at $q = 4$ and $q = 11$, and the stated congruence leaves only $q = 11$. Conversely, Clebsch hexagons exist over $\mathbb{F}_{11}$ by Dye's Theorem 1, and the inclusion above is then an equality because both sets have twelve points. $\square$

The theorem is an unconditional field-size classification with a conceptual small-field proof. Within the icosahedral construction, three further features meet at 11: $p + 1 = 12$, so the twelve vertices of the icosahedron exhaust the projective line; p is prime to 60, so the icosahedral group survives reduction faithfully; and the fifteen secants of a six-arc have enough capacity to cover every point off the conic. The failure of the same symmetric recipe at the next admissible prime is nevertheless worth recording concretely. At $q = 19$ ($19 \equiv -1 \pmod{10}$, so the rationality filter passes and only Lemma 6.1 excludes it) the same recipe — the poles of the six five-fold axes of an icosahedral $A_5 < \text{PSL}_2(19)$ — does still produce a genuine six-arc disjoint from the conic, so the object exists; what fails is the covering. Proposition 6.4 now gives directly

$$|\mathcal{U}(A)| = 140 = 20 + 120$$

against a conic of only $q + 1 = 20$ points. An independent finite replay reconstructs the arc and reproduces this value. The arithmetic differs from $q = 11$ in a way worth noting: there $5 \mid q - 1$, so an order-five element splits and fixes two rational points of $\mathcal{C}$, whereas at $q = 19$ we have $5 \mid q + 1$, so the order-five elements lie in a non-split torus and fix no rational conic point at all — their fixed pairs are conjugate over $\mathbb{F}_{361}$. The chord through such a pair is Galois-stable and hence still a rational line, so its pole is still a rational point and the arc is still defined; we verify this in the script rather than assume it.

The shape of the failure is instructive. The twenty conic directions remain in the projective deep-hole syndrome locus — $\mathcal{C} \subset \mathcal{U}(A)$ still — by the same classical edge criterion [16, p. 281], but they no longer exhaust $\mathcal{U}(A)$: $120 = (19 - 4)(19 - 11)$ further points of $\text{PG}(2, 19)$ escape all fifteen secants, and $\mathcal{U}(A)$ lies on no conic (the quadratic-monomial evaluation matrix of $\mathcal{U}(A)$ has full rank six). The earlier capacity bound from Lemma 6.1 predicts only that at least 85 off-conic points escape; the Clebsch-family formula gives the exact excess 120. At $q = 11$ the same capacity, 150, comfortably exceeds the 115 points needing cover, and the excess is spent on overlap. The Clebsch objects therefore persist, and the associated conic remains inside the projective syndrome locus when $q \equiv 3 \pmod{4}$, but the exact filling is isolated at $q = 11$ by the factorization above.

6.1 The small-arc boundary

The chord moments also put the six-point theorem into a short classification for all smaller neighboring arc sizes.

Theorem 6.5 (Conic-filling loci for $4 \leq k \leq 7$). *Let A be a k -arc in $\text{PG}(2, q)$, where q is a prime power and $4 \leq k \leq 7$. If $\mathcal{U}(A)$ is the full set of $\mathbb{F}_q$ -rational points of a nonsingular conic, then*

$$(k, q) = (4, 5) \quad \text{or} \quad (k, q) = (6, 11).$$

In the first case A is a projective four-frame; in the second it is a Clebsch hexagon. Conversely, both cases occur.

Proof with exhaustive finite verification. Let n_i count off-arc points incident with exactly i chords. For $k \leq 7$, at most three pairwise disjoint chords concur. Counting chord–point incidences and pairs of disjoint chords gives the two universal moments

$$\sum_i i n_i = \binom{k}{2} (q - 1), \quad \sum_i \binom{i}{2} n_i = 3 \binom{k}{4}.$$

For $k = 4$, the second moment forces $n_2 = 3$, and hence $|\mathcal{U}(A)| = (q - 2)(q - 3)$. Equating this with $q + 1$ gives $q = 5$. Every four-arc is a projective frame, and for the standard frame a direct substitution gives

$$\mathcal{U}(A) = Z(X^2 + Y^2 + Z^2 + XY + XZ + YZ),$$

a nonsingular six-point conic. For $k = 5$, the same moments force $n_2 = 15$ and $|\mathcal{U}(A)| = q^2 - 9q + 21$; equality with $q + 1$ would require $q^2 - 10q + 20 = 0$, which has no integral root. The case $k = 6$ is Theorem 6.3 together with Theorem 4.4.

For $k = 7$, the moments give

$$|\mathcal{U}(A)| = q^2 - 20q + 120 - n_3.$$

If this equals $q + 1$, then

$$n_3 = q^2 - 21q + 119, \quad n_2 = 105 - 3n_3, \quad n_1 = 3(q^2 - 14q + 42).$$

The standard arc-size bound gives $q \geq 7$. The condition $n_2 \geq 0$ gives $q \leq 15$, so the prime-power candidates are $q \in \{7, 8, 9, 11, 13\}$; then $n_1 \geq 0$ eliminates 7, 8, 9. Thus only $q = 11$ and $q = 13$ remain, with forced spectra $(n_1, n_2, n_3) = (27, 78, 9)$ and $(87, 60, 15)$ respectively. An exhaustive four-frame-normalized enumeration finds no conic-filling seven-arc in either plane. Every seven-arc is reached because it contains a four-frame; equivalently, the checker

extends every normalized six-arc by every point of its uncovered locus and verifies that each resulting seven-arc occurs exactly three times, according to which non-frame point is deleted. For all candidates with $|\mathcal{U}(A)| = q + 1$, the quadratic evaluation matrix has full rank six. The finite replay checks every leaf, including the $q = 5$ equality; the universal moment reductions are kernel-checked. $\square$

The $q = 5$ four-frame equality is elementary and is stated here without an independent priority claim. Dye’s adjacent $q = 5$ degeneration concerns the six points on a conic and their ten internal triple-chord points, not this four-frame uncovered-locus statement [16, §1.4]. The role of the theorem here is to give the unified range $4 \leq k \leq 7$; its new certified component is the exclusion of both surviving seven-arc fields. For $k \geq 8$ the second moment leaves additional concurrency parameters, so the analogous classification remains open.

7 Verification architecture and outlook

Table 1 separates cited input, kernel-checked finite certificates, and fail-closed enumeration. Lean and script paths are relative to `lean/RelativeConicArcs/` and the paper directory; commands, hashes, sentinels, and import closures will accompany the archive.

Result	Cited or proved input	Lean certificate	Executable replay
A_5 orbits and syndrome conic	Dye; Proposition 3.1	<code>Q11A5PointOrbits.lean</code> ; <code>Q11Coding.lean</code>	<code>check_rigidity_degenerate_conic.py</code>
Decoding and chirality	arc-coset dictionary; Edge-Dye triangles	<code>Q11DecodingSynthesis.lean</code>	<code>check_decoding.py</code> ; <code>check_chirality.py</code> ; <code>check_code_automorphisms.py</code>
Rigidity theorem	line bound; chord defect; Dye’s Theorem 1	<code>Q11DyeAxioms.lean</code> ; <code>Q11DyeConsequences.lean</code> ; <code>SixArcDefectBridge.lean</code>	1548-representative rigidity replay
Gap and low-degree loci	finite linear algebra over $\mathbb{F}_{11}$	—	<code>check_global_conic_gap.py</code> ; <code>check_perturbation_gap.py</code> ; <code>check_low_degree_loci.py</code> ; Singular replay
Uniqueness of $q = 11$	chord moments; cited Sylvester bound	<code>ClebschChordDefect.lean</code> ; <code>Q9Sylvester.lean</code>	<code>check_small_q_uniqueness.py</code> ; from-scratch $q = 9$ certificate
Clebsch-family formula	Dye’s ten Brianchon points and edge criterion	—	<code>check_q19_nonexample.py</code>
The range $4 \leq k \leq 7$	two universal chord moments	<code>SmallKChordMoments.lean</code> ; <code>SmallKGeometricBridge.lean</code>	<code>check_small_k_conic_filling.py</code>

Table 1: Trust and verification map for the principal results.

Open directions are higher secant-index moments for $k \geq 8$, exhaustion of the degree-six kernels from Remark 4.7, and the transition from fixed- k classifications to the large-arc conic-forcing regime. The last problem first requires verified thresholds and field hypotheses.

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